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Document Owner: Principal Enterprise Architect

Reference Architecture Library: VI Business (India)

1) Executive Purpose and Applicability

The Reference Architecture Library maintains VI Business (India)'s pre-certified network and security topologies. Adhering to these documented architectural bounds structurally mathematically guarantees deployments satisfy contracted SLAs while minimizing deviation risk during Zero-Touch Provisioning (ZTP) escalations.

Intended Audience: Solution Architects, Pre-Sales Engineers, Technical RFP Responders. **Applicability:** Designed to be directly replicated into RFP technical volumes as network diagrams and topological justifications supporting compliance thresholds.

2) Definitions and Assumptions

Definitions:

- **ZTP:** Zero-Touch Provisioning.
- **ECMP:** Equal-Cost Multi-Path routing.
- **cAPN:** Custom Access Point Name.

Assumptions:

- **Internal Control:** Submitting "custom" architectural shapes beyond the bounds of these defined templates requires manual review from the Principal Enterprise Architect. Unvetted architectures void standard SLA penalty guarantees.

3) Detailed Core Sections

3.1 Tier-1 Hub Architecture (The 99.99% Availability Model)

- **Topology Base:** High-Availability (HA) clustered chassis edge nodes.
- **Underlay Fabric:** Primary 100+ Mbps Dedicated Internet Access (DIA) Fibre // Secondary isolated 4G/5G LTE macro.
- **Routing Logic:** BGP (Border Gateway Protocol) dynamically propagating paths across both legs. Executes Active/Active transport utilizing ECMP.
- **Application-Aware Logic:** Deep packet shaping policies recognize latency-intolerant payloads (e.g., SAP S/4HANA instances) routing them exclusively down the lowest-jitter path. Capable of executing stateful sub-50ms failover without destroying the application handshake session.

3.2 Tier-2/3 Branch Architecture (The Mass-Rollout Model)

- **Topology Base:** Single discrete SD-WAN Edge/Next-Generation Firewall (NGFW) appliance per location.

- **Underlay Fabric:** Broadband FTTx (Primary) with robust Business LTE (Backup/Failover).
- **Deploy Logic:** Explicitly orchestrated via Zero-Touch Provisioning (ZTP). Hardware bootstraps instantly, reaching out to the central controller (hosted physically inside India) to download site-specific IP/routing templates, achieving operational live-status `< 4 hours` from hardware unboxing.

3.3 Data Center to Hyperscaler Interconnect Architecture

- **Topology Base:** Dedicated dual 10 Gbps Layer 2/3 circuits originating directly from VI's NESCO IT Park (Mumbai) or Chennai Tier-III nodes bridging into the Hyperscaler Meet-Me-Room (MMR).
- **Azure ExpressRoute:** Implements Active-Active resilient eBGP, anchoring in Azure Central India (Pune) to support dedicated Microsoft and Private Peerings.
- **AWS Direct Connect:** Specialized integration pipelines terminating into high-availability AWS Transit Gateways mapped to the ap-south-1 region.
- **Performance:** Maintains sub- `8ms` round trip base-latency independent of public ISPs.

3.4 Enterprise IoT / EMM Traffic Architecture

- **Topology Base:** Fleet sensors communicate across VI's nationwide macro RAN base (250,000+ cell sites), terminating in dynamically isolated Packet Data Network Gateways (PGW).
- **Routing Logic:** Traffic flows exclusively into heavily localized Custom APNs (cAPN). Consequently, IoT payload is permanently encrypted and separated from the wider internet grid. The cAPN directly intersects the corporate network backbone via localized private IP addressing.

4) Evidence and Proof-Artifact Map

Architecture Type	Source `.vsdx` Visio Artifact	Owner
Tier-1 Active-Active HA	`SharePoint/VIB_Arch/Tier1_Dual_CPE.vsdx`	Core Architecture Desk
NESCO to Azure ExpressRoute	`SharePoint/VIB_Arch/Azure_ER_Logical.vsdx`	Cloud Solutions Architect

5) Governance and Ownership

- **Owner:** Principal Enterprise Architect
- **Reviewers:** VP Core Networks, Head of Managed Services, Enterprise CISO.
- **Review Frequency:** Updated concurrently matching major OEM firmware capability releases or yearly network infrastructure refreshes.

6) Constraints, Exclusions, and Dependencies

- **Design Constraints:** Proposing the 99.99% Tier-1 SLA topology mathematically requires the customer to provide dual, distinct power phases backed by independent UPS units on-site. The SLA guarantee structurally collapses if localized power supply is unstable.
- **DPDP Compliance Dependency:** All architectures depicted utilize exclusively Indian-operated management planes to satisfy the strict Data Localization limits of the Digital Personal Data Protection Act 2023.